Capacitor

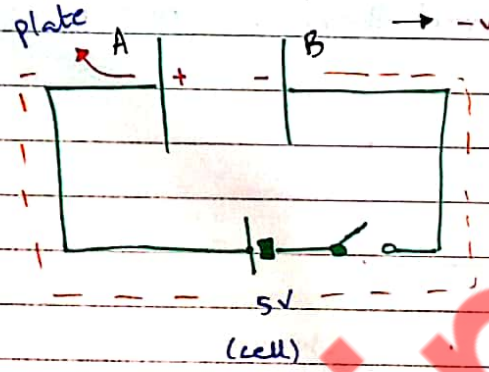
A device which is used to store the charge

(OR)

A device which is used to store electric energy

→ Magnitude of charge depends on electric potential
→ If charge of plates is zero then electric potential will be zero

Parallel Plate Capacitor

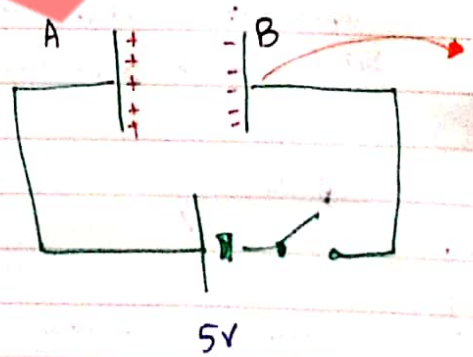


→ -ve charge goes back to positive terminal of cell

→ Initially cell is at higher potential and plates are at lower potential

→ when switch is closed, charge travels from cell to plates

→ Only negative charge moves (not positive charge)
→ when switch closed, negative charge from cell moves to plate B and plate B becomes a charged conductor.
→ Plate A now becomes charged, inner surface of plate A is positively charged while inner surface of plate B is negatively charged



First charge, since it has charge, it's electric potential difference would be 1V

Aamir Yasin

PAPERWORK
Scanned with CamScanner

→ After first charge [illegible] charges arrive at plate surface and charge increases and so does p.d

→ At the end the final potential difference of plates also becomes 5V as 5 charges arrive at its surface.

Potential difference across capacitor = V

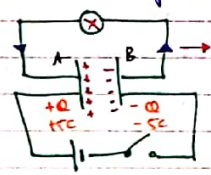
→ Equal charge stored by plate A of capacitor as of plate B of same capacitor

↳ If B has −Q charge stored then A would have +Q charge stored

Total charge stored in capacitor = Q

* How do we know that total charge stored is Q;

1- Open the switch so capacitor is not joined with cell
2- Connect both plates with metal wire and connect bulb in it.
3- Charge stored by plate B = −5Q
   Charge stored by plate A = +5Q
4-



→ Move towards A

→ when joined by metal rod, negative charge moves towards positive charge and cancel them causing total charge to be Q.

In capacitor,
potential difference = V
Charge = Q

$Q \propto V$ → $\boxed{Q = CV}$ → $C = \frac{Q}{V}$

Q : constant (V)
↓
capacitance is constant

Farad = $\frac{\text{Coulomb}}{\text{Volt}}$

Capacitance

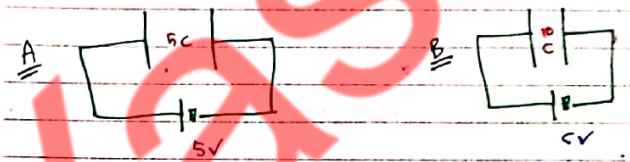
Charge to potential difference ratio for capacitor is called "Capacitance"

(OR)

Ability of capacitor to store charge

Farad

Capacitance of a capacitor is one farad if it stores a charge of one coulomb under the potential difference of 1 volt.



→ Both A & B are connected to identical cell of 5V

→ A has 1 farad capacitance as ratio of coulomb to volt is same
→ B has 2 farad capacitance as 10 charge is stored for the voltage
→ B stores more charge as capacitance of capacitor is also affected by area and distance between the plates

$C \propto A$ —i

$C \propto \frac{1}{d}$ —ii

→ d is distance between the plates
→ ε_0 here is constant for vacuum and air medium

$C \propto \frac{A}{d}$ $C = (\text{constant})\frac{A}{d}$

$$\boxed{C = \varepsilon_0 \frac{A}{d}}$$

→ This expression is for only parallel plate capacitor

$$\boxed{C = \frac{Q}{V}}$$

→ It is for general capacitor (Can be used for every capacitor)

Q - Total charge stored by capacitor
V - Potential difference across capacitor

PAPERWORK

Aamir Yasin

Scanned with CamScanner

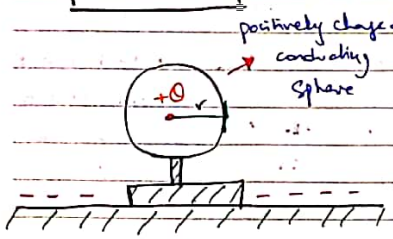
3-12-21 Friday

Capacitor

$$C = \frac{Q}{V}$$ — for any type of capacitor

Q - Total charge stored by capacitor
V - Total p.d across the capacitor

$$C = \varepsilon_0 \frac{A}{d}$$ — for any type of parallel capacitor



positively charged conducting sphere

→ For a capacitor we need two oppositely charged conductors

→ The positive conducting sphere induces negative charge at the bottom thus the arrangement showcases a "Capacitor".

↳ It's not a parallel plate capacitor

* Charge = Q
Radius = r

Derivation

$$C = \frac{Q}{V}$$

→ if potential at surface of sphere =>

$$V = \frac{kQ}{r}$$

$$= Q \div \frac{kQ}{r}$$

→ earth potential is "zero"

$$= \cancel{Q} \times \frac{r}{k\cancel{Q}}$$

→ to find potential difference "$\frac{kQ}{r} - 0$"

$$= \frac{r}{k} \Rightarrow \frac{1}{k} \times r$$

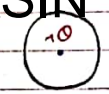
$$K = \frac{1}{4\pi\varepsilon_0}$$

$$\frac{1}{k} = 4\pi\varepsilon_0$$

substitute,

$$C = (4\pi\varepsilon_0) r$$ — for any charged sphere only

positively charged conductor



Total charge is concentrated at the centre of the sphere

$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/Nm^2$ $V = 4 \times 10^5\ V$
$r = 36cm$

i) C = ?
ii) Q = ?

Solution

$$C = (4\pi\varepsilon_0) r$$

$$= (4 \times 3.14 \times 8.85 \times 10^{-12}) \times 0.36$$

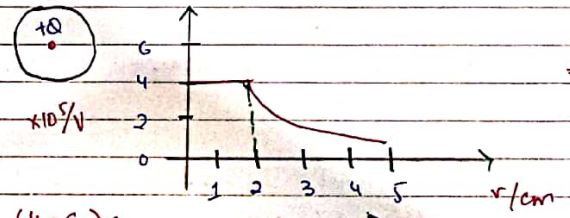
$$C = 4.0 \times 10^{-11}\ F$$

$$Q = CV$$

$$= 4.0 \times 10^{-11} \times 4 \times 10^5$$

$$Q = 1.6 \times 10^{-5}\ C$$

Q.2



=> r = 2cm as constant line shows the sphere (internal)

a) $C = (4\pi\varepsilon_0) r$
b) $Q = CV$

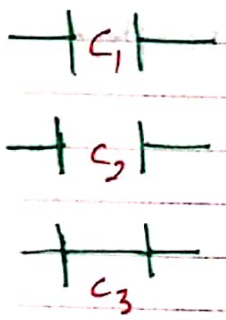
Solution

Aamir Yasin

PAPERWORK

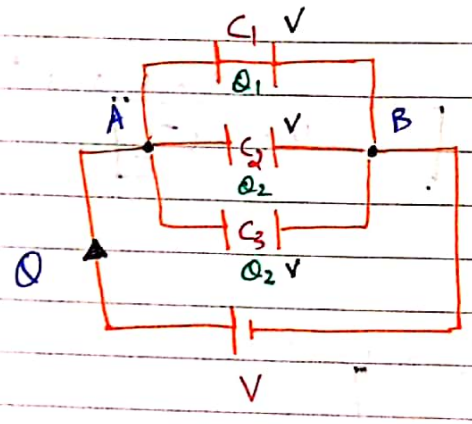
Scanned with CamScanner

Parallel Combination of Capacitors



→ All 3 capacitors have different capacitance

→ If an electrical component is directly connected to the battery/cell so $\boxed{P.d = e.m.f}$



=> The "Q" split at A and different for C_1, C_2 and C_3.

Properties

In parallel combination

1- Potential difference across all the capacitors is the same and its equal to the e.m.f of cell

→ thus charge stored by each capacitor is different
↳ cz capacitance is different

2- Charge stored by each capacitor is different and magnitude of charge stored by the capacitor is directly dependent on the capacitance of the capacitor

3- Sum of charge stored by all the capacitors is equal to total charge delivered by the cell.

$$\boxed{Q = Q_1 + Q_2 + Q_3}$$

Aamir Yasin

Scanned with CamScanner

$Q = CV$

$Q_1 = C_1V$

$Q_2 = C_2V$

$Q_3 = C_3V$

$$Q = Q_1 + Q_2 + Q_3$$

$$CV = C_1V + C_2V + C_3V$$

$$C\cancel{V} = (C_1 + C_2 + C_3)\cancel{V}$$

$$\boxed{C = C_1 + C_2 + C_3}$$ — 4th property & derivation is in syllabus

Effective capacitance

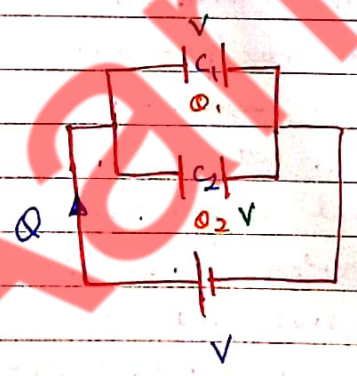
=> Net or Resultant capacitance of parallel/any combination

4- Effective capacitance is equal to the sum of all capacitance of all the capacitors.

5- Effective capacitance is greater than the greatest capacitance of the combination

↳ We form parallel combination of capacitors to increase the capacitance of the arrangement

Direct Method



$$\boxed{C = C_1 + C_2}$$

$Q = CV$

$Q_1 = C_1V$ —— (1)

$Q_2 = C_2V$ —— (2)

$Q = CV$

$Q = (C_1 + C_2)V$

$$V = \frac{Q}{C_1 + C_2}$$

so (on next page)

PAPERWORK

Scanned with CamScanner

$$Q_1 = C_1 \times \frac{Q_1}{C_1 + C_2}$$

$$\boxed{Q_1 = \frac{C_1}{C_1 + C_2} \times Q}$$ — derivation in syllabus

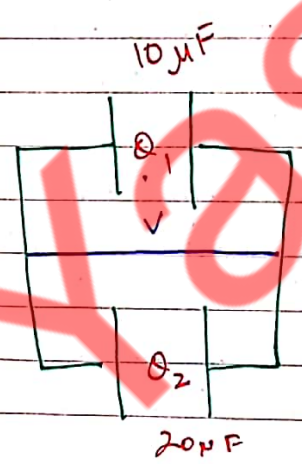
$$\boxed{Q_2 = \frac{C_2}{C_1 + C_2} \times Q}$$

(Indirectly a ratio method & only if 2 capacitors connected in parallel)

Q.

a)

b)



$Q = ?$

$Q = C \times V$

$= 10 \times 30$

$$\boxed{Q = 300\,\mu C}$$

After charging fully, the capacitor in (a) is connected with a capacitor of 20 μF

→ 10 μF being a charged capacitor has max potential while 20 μF capacitor being uncharged will have "0" potential so potential is transfered from 10 μF to 20 μF until it becomes the same

i- Q_1 & Q_2 ?

ii- V ?

Aamir Yasin

Scanned with CamScanner

Solution

$$Q_1 = \left(\frac{C_1}{C_1+C_2}\right)Q$$

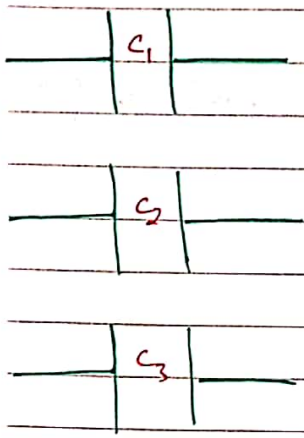
$$Q_2 = \left(\frac{C_2}{C_1+C_2}\right)Q$$

ii - $Q = CV$

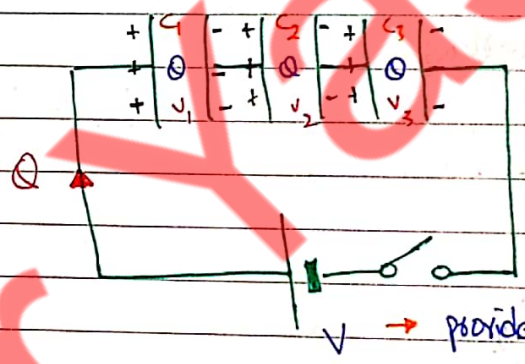
$$V = \frac{Q}{C}$$

$$= \frac{300\,\mu}{10\,\mu}$$

$$\boxed{V = 10\,V}$$



Series Combination



→ provides $3e^-$ → every capacitor get's $3e^-$ charge

Properties

1- Charge stored by each capacitor is the same and is equal to total charge delivered by the cell $V = \frac{Q}{C}$

2- Potential difference across each capacitor is different and magnitude of potential difference across the capacitor inversly depends upon capacitence of the capacitor

3- Sum of potential difference across all the capacitor is equal to the e.m.f of cell

$$\boxed{V = V_1 + V_2 + V_3}$$

$$V_1 = \frac{Q}{C_1} \qquad V_2 = \frac{Q}{C_2} \qquad V_3 = \frac{Q}{C_3}$$

Aamir Yasin

PAPERWORK

Scanned with CamScanner

$V = V_1 + V_2 + V_3$

$\frac{Q}{C} = \frac{Q}{C_1} + \frac{Q}{C_2} + \frac{Q}{C_3}$

$\frac{Q}{C} = Q\left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}\right)$

$$\boxed{\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}}$$

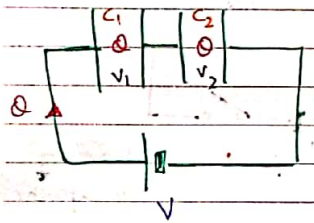
→ derivation is in syllabus

4- Reciprocal of effective capacitance is equal to the sum of reciprocals of capacitence of all the capacitors

5- Effective capacitence is less than the least capacitance of the combination

↳ series to decrease the capacitance

Direct Method



$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}$$

$$\frac{1}{C} = \frac{C_1 + C_2}{(C_1)(C_2)}$$

$$C = \frac{(C_1)(C_2)}{C_1 + C_2}$$

$$\boxed{V_1 = \left(\frac{C_2}{C_1 + C_2}\right)V}$$

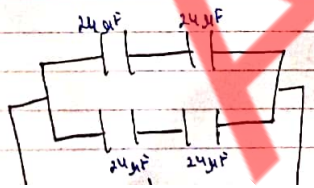
$$\boxed{V_2 = \left(\frac{C_1}{C_1 + C_2}\right)V}$$

$$\boxed{\text{Effective capacitence} = \frac{\text{Product}}{\text{Sum}}}$$

Q1) 48 μF, 25V (number of capacitors, each 48 μF, solve working voltage 25V)

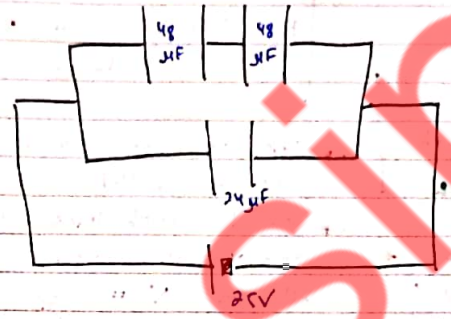
↳ we have to make the arrangements

a) 48 μF, 50V



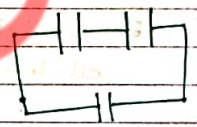
→ as in series C is product and in parallel it's sum addition is that's why 48 μF and 25V each.

b) 72 μF, 25V

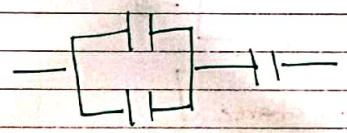


Q-2 6 μF = 3 capacitors

a) 9 μF



b) 4 μF

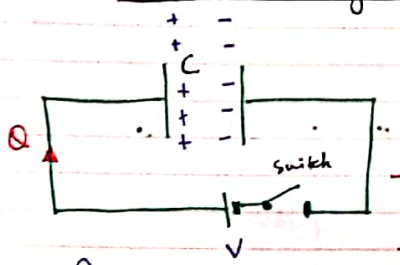


(on next page)

Aamir Yasin

PAPERWORK
Scanned with CamScanner

Energy stored by the capacitor



$$\boxed{E.M.F = \frac{\text{Energy delivered}}{\text{charge}}}$$

→ capacitor is connected with a cell via a switch

→ Once the switch closes, the capacitor gets charged by the cell thus $V = \frac{\text{energy delivered}}{\text{(ideally)}}$

stores electrical energy (given from cell)

charge = Q

potential difference = V

Capacitance = C

$Q = CV$

total charge delivered by cell

$V = \frac{Q}{C}$

Energy delivered by cell to capacitor = charge × E.M.F

$= QV$

$= (CV) \times V$

$= CV^2$

(OR) QV, $Q \times \frac{Q}{C}$, $\frac{Q^2}{C}$

Energy delivered by the cell to capacitor

$\boxed{QV}$ — ①

$\boxed{CV^2}$ — ②

$\boxed{\frac{Q^2}{C}}$ — ③

$$\boxed{\text{Energy Stored} = \left(\frac{\text{Energy}}{\text{charge}}\right) \times \text{total charge}}$$

$= \left(\frac{V+0}{2}\right) \times Q$

→ V is the energy/charge delivered to each charge but not all charges stored e.m.f equal to V

→ If we find out the energy stored by each charge and the total charge stored we can find the total energy stored by the capacitor

$\boxed{E = \frac{1}{2}QV}$ ①

$\boxed{E = \frac{1}{2}CV^2}$ ②

$\boxed{E = \frac{1}{2}\frac{Q^2}{C}}$ ③

volt = $\frac{\text{Joule}}{\text{coloumb}}$

∴ $5V = \frac{5J}{1C}$

→ Energy provided by cell to each charge is equal to V

→ We assume the charge is uniformly distributed we assume all electrons have 1C charge — cell provides each e^- with 5J

→ Once the switch is closed it has 5J of energy is taken by e^- from cell into the capacitor

→ Once the second $\bar{e}$ is emitted with 5J of energy from cell due to repulsion b/w the 2 $\bar{e}$s it loses energy.

→ The same process is followed by each electron emitted

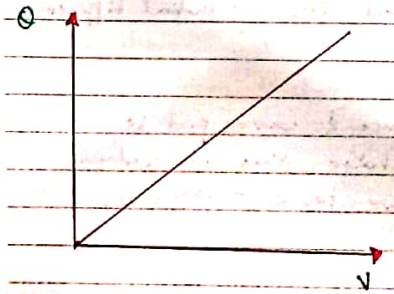
→ First $\bar{e}$ has maximum possible energy and the last $\bar{e}$ with minimum possible of energy — ~~some~~ not all $\bar{e}$s store same energy

Max possible energy = V

Min " " = 0 ⇒ so avrge energy stored by each charge of capacitor

$= \frac{V+0}{2}$

→ Capacitor stores half of energy ~~stored~~ provided by the cell as half of energy is wasted in the charging process



→ constant

$Q = CV$

$Q \propto V$

(that's why straight line)

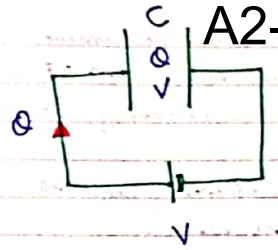
→ Area of the graph = $\frac{1}{2}QV$ or Energy Stored

(on next page)

Aamir Yasin

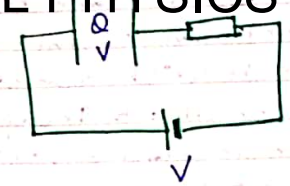
PAPERWORI

Scanned with CamScanner



Energy delivered $= QV$

Energy stored $= \frac{1}{2}QV$



→ Same cell & capacitor but is connected in series with a resistance

Energy delivered $= QV$

Energy stored $= \frac{1}{2}QV$

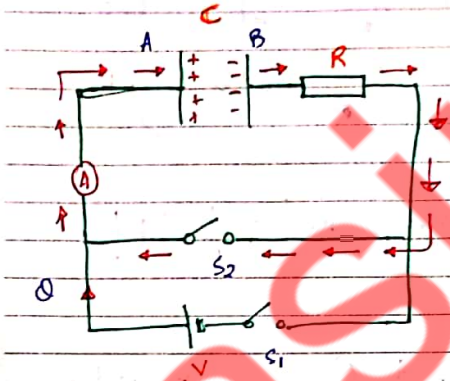
↳ It remains same even though a resistor is connected

→ When capacitor is fully charged it's potential difference will be V (e.m.f of cell)

→ $Q = CV$, if same capacitance & same p.d so same charge thus all values same therefore same expressions.

→ The only difference will be that more time will be taken for the one with resistor

Discharging of a capacitor



$$I = -\frac{\Delta Q}{\Delta t}$$

During Discharge

↳ −ve sign shows that charge of capacitor is decreasing

→ If S_2 ~~closed~~ open and S_1 closed, then the capacitor will be charged

↳ p.d across the capacitor = V

→ Now we open S_1 and close S_2

→ To discharge a capacitor, remove it's connection with the cell and join the plates together.

→ Now charge will flow from B to A, so ~~that~~ until the charge is being discharged there will be flow of current through the ammeter

→ There will be change in charge during discharge, represented by −ve or +ve sign as −ve sign − decrease in charge
+ve sign − increase in charge

$$I = -\frac{\Delta Q}{\Delta t}$$

$$\Delta Q = -I\Delta t$$

$$\Delta Q = -\frac{V}{R}\Delta t \quad \because I = \frac{V}{R}$$

$$\Delta Q = -\frac{Q}{RC}\Delta t \quad \because V = \frac{Q}{C}$$

$$\frac{\Delta Q}{Q} = -\frac{1}{RC}\Delta t$$

$$\int \frac{\Delta Q}{Q} = \int -\frac{1}{RC}\Delta t$$

Symbol of Integration

mathematical function

PAPERWORK

Aamir Yasin

Scanned with CamScanner

$\int \frac{\Delta Q}{Q} = \boxed{-\frac{1}{RC}} \int \Delta t$ → putting limits

Q_0 at $t=0$

Q : after $t=t$

constant factor

→ We put upper limit first & then lower limit

$\left| \ln Q \right|_{Q_0}^{Q} = -\frac{1}{RC} \left| t \right|_{0}^{t}$

$\ln Q - \ln Q_0 = -\frac{1}{RC}(t-0)$ → shift + ln to get inverse ln

$\ln\left(\frac{Q}{Q_0}\right) = -\frac{1}{RC} t$

$\frac{Q}{Q_0} = e^{-\frac{t}{RC}}$ → exponential (reciprocal of ln)

$e = \frac{1}{\ln}$

$$\boxed{Q = Q_0 e^{-\frac{t}{RC}}}$$

Q - charge left after given time
Q_0 - Initial charge of capacitor
t - time
R - Resistance of resistance
C - capacitance of capacitor

$$\boxed{I = I_0 e^{-\frac{t}{RC}}}$$ → Discharging current at any instant

I_0 - Initial current

$$\boxed{V = V_0 e^{-\frac{t}{RC}}}$$ → Discharging potential difference across the capacitor

V_0 - Initial p.d across the capacitor

→ $Q = Q_0 e^{-\frac{t}{RC}}$

↳ $Q = Q_0 e^{-\frac{RC}{RC}}$ if we replace t by RC

t = RC - represents time

$Q = Q_0 e^{-1}$

$Q = Q \times \frac{1}{e}$

$Q = \frac{1}{e} Q_0$

$$\boxed{Q = 0.37 Q_0}$$

Time Constant → RC (also called "time constant")

Time after which charge stored by the capacitor reduces to $0.37 Q_0$ of its initial charge

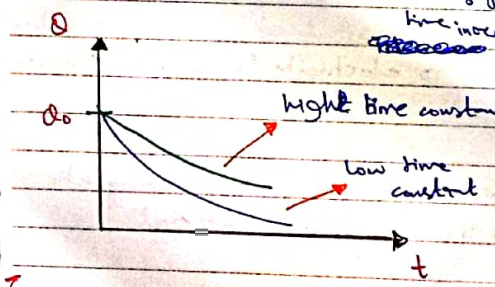
$$\boxed{T = RC}$$

to increase T we will increase R or C or both

↳ effect on graph (shown by green line in graph)

discharging time increases

For Discharging

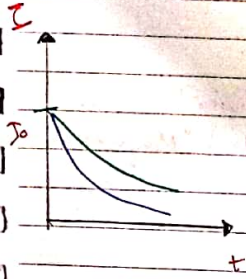


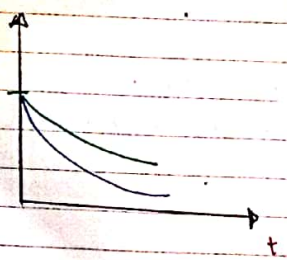
=> Exponential graph

never touches x-axis

so

same graph for I & V with t





Aamir Yasin

Scanned with CamScanner